

18. Write a Pandas program to split the following given dataframe into groups based on school code and class.

CODE:

```
import pandas as pd
```

```
data = {  
    'school_code': ['s001', 's001', 's002', 's002', 's003', 's003'],  
    'class': ['V', 'VI', 'V', 'VI', 'V', 'VI'],  
    'student_name': ['Raj', 'Priya', 'Arun', 'Meena', 'Kiran', 'Vijay'],  
    'age': [10, 11, 10, 12, 11, 12]  
}
```

```
df = pd.DataFrame(data)
```

```
grouped = df.groupby(['school_code', 'class'])
```

```
for name, group in grouped:
```

```
    print("\nGroup:", name)
```

```
    print(group)
```

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```
import pandas as pd  
  
data = {  
    'school_code': ['s001', 's001', 's002', 's002', 's003', 's003'],  
    'class': ['V', 'VI', 'V', 'VI', 'V', 'VI'],  
    'student_name': ['Raj', 'Priya', 'Arun', 'Meena', 'Kiran', 'Vijay'],  
    'age': [10, 11, 10, 12, 11, 12]  
}  
  
df = pd.DataFrame(data)  
  
grouped = df.groupby(['school_code', 'class'])  
  
for name, group in grouped:  
    print("\nGroup:", name)  
    print(group)
```

```
Group: ('s001', 'V')  
  school_code class student_name age  
0      s001      V          Raj    10  
  
Group: ('s001', 'VI')  
  school_code class student_name age  
1      s001      VI          Priya  11  
  
Group: ('s002', 'V')  
  school_code class student_name age  
2      s002      V          Arun    10  
  
Group: ('s002', 'VI')  
  school_code class student_name age  
3      s002      VI          Meena  12  
  
Group: ('s003', 'V')  
  school_code class student_name age  
4      s003      V          Kiran   11  
  
Group: ('s003', 'VI')  
  school_code class student_name age  
5      s003      VI          Vijay  12
```

19. Write a Pandas program to display the dimensions or shape of the World alcohol consumption dataset. Also extract the column names from the dataset.

Code:

```
import pandas as pd
```

```
data = {  
    'Year': [1987, 1989, 1985, 1986, 1987],  
    'WHOregion': ['Africa', 'Americas', 'Africa', 'Europe', 'Asia'],  
    'Country': ['Algeria', 'Brazil', 'Nigeria', 'France', 'India'],  
    'Beverage Types': ['Wine', 'Beer', 'Spirits', 'Wine', 'Beer'],  
    'Display Value': [0.1, 2.5, 0.3, 8.7, 1.2]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Shape of Dataset:")
```

```
print(df.shape)
```

```
print("\nColumn Names:")
```

```
print(df.columns)
```

```
File Edit Format Run Options Window Help  
import pandas as pd  
  
data = {  
    'Year': [1987, 1989, 1985, 1986, 1987],  
    'WHOregion': ['Africa', 'Americas', 'Africa', 'Europe', 'Asia'],  
    'Country': ['Algeria', 'Brazil', 'Nigeria', 'France', 'India'],  
    'Beverage Types': ['Wine', 'Beer', 'Spirits', 'Wine', 'Beer'],  
    'Display Value': [0.1, 2.5, 0.3, 8.7, 1.2]  
}  
  
df = pd.DataFrame(data)  
  
print("Shape of Dataset:")  
print(df.shape)  
  
print("\nColumn Names:")  
print(df.columns)
```

```
Shape of Dataset:  
(5, 5)
```

```
Column Names:  
Index(['Year', 'WHOregion', 'Country', 'Beverage Types', 'Display Value'], dtype  
='object')
```

20. Write a Pandas program to find the index of a given substring of a DataFrame column.

Code:

```
import pandas as pd
```

```
df = pd.DataFrame({  
    'Name': ['Raj Kumar', 'Priya Sharma', 'Arun Kumar', 'Meena Singh']  
})
```

```
substring = 'Kumar'
```

```
df['Index_Position'] = df['Name'].str.find(substring)
```

```
print(df)
```

```
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```

```
import pandas as pd
```

```
df = pd.DataFrame({  
    'Name': ['Raj Kumar', 'Priya Sharma', 'Arun Kumar', 'Meena Singh']  
})
```

```
substring = 'Kumar'
```

```
df['Index_Position'] = df['Name'].str.find(substring)
```

```
print(df)
```

```

Name      Index_Position
0      Raj Kumar          4
1  Priya Sharma          -1
2      Arun Kumar          5
3  Meena Singh           -1
|

```